

Exact Domination of Lecture Timings

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Code: <https://github.com/deveshk-boop/daa-project-q17> (build and run instructions are in the README).

1 The problem

We have n lectures, one per day, each with a start and end time (possibly crossing midnight). We must pick a subset S of lectures so that every lecture overlaps *exactly one* lecture of S ; a chosen lecture counts as overlapping itself. If no such subset exists, we say so.

Setup. Times are in minutes on a circle of length $M = 1440$. Lecture i is the arc A_i of $\ell_i = (e_i - s_i) \bmod M$ minutes starting at s_i . I made two assumptions since the statement is silent on them: lectures are half-open, so back-to-back lectures do not overlap, and a lecture with $s_i = e_i$ lasts one minute. Write $d(x, y) = (y - x) \bmod M$ for the clockwise distance from x to y ; a time p is in A_i iff $d(s_i, p) < \ell_i$.

2 Key observations

Overlap test. Two arcs overlap iff one of them starts inside the other. (If p is in both, the one whose start is closer behind p starts inside the other.) So the overlap table takes $O(1)$ per pair.

Chosen lectures are disjoint. A chosen lecture already overlaps itself, so it may not overlap any other chosen lecture.

Single lecture. $S = \{a\}$ is valid iff a overlaps every lecture.

Two or more lectures. Then S is a set of disjoint arcs, so they sit around the circle in a fixed order $x_0, x_1, \dots, x_{k-1}$. For two neighbours a then b , the *gap* is the time between the end of a and the start of b . Say b can follow a , written $F(a, b)$, if:

1. a and b do not overlap;
2. no lecture overlaps both a and b (it would be covered twice);
3. no lecture that overlaps neither a nor b starts inside the gap (it would not be covered at all).

Claim. A set S of $k \geq 2$ disjoint arcs is valid iff $F(x_i, x_{i+1})$ holds for every neighbouring pair (cyclically, including $x_{k-1} \rightarrow x_0$).

Proof. ($\Rightarrow$) If S is valid, condition 1 holds because the arcs are disjoint. Condition 2 holds, otherwise some lecture is covered twice. For condition 3, suppose c overlaps neither a nor b but starts in their gap. It cannot reach the start of b , so it lies entirely inside the gap. No chosen arc touches the gap, because a and b are neighbours, so c overlaps nothing in S , a contradiction.

($\Leftarrow$) Take any lecture c . If c overlapped two chosen arcs, then, since c is one connected stretch of the circle, walking from one of them to the other it must touch a pair of neighbours, and condition 2 forbids that. If c overlapped no chosen arc, its start would fall in some gap, and it would overlap neither neighbour of that gap, which condition 3 forbids. So c overlaps exactly one chosen arc. $\square$

3 Algorithm

1. Build the overlap table.
2. If some lecture overlaps all lectures, output it.
3. Otherwise build the table $F(a, b)$ for all pairs, by checking the three conditions against every lecture c .
4. Lecture 1 must overlap some chosen lecture, so try each lecture a_0 that overlaps lecture 1 as the starting point. For each one:
 - Sort the other lectures by $\text{pos}(x) = d(s_{a_0}, s_x)$, the clockwise offset of their start from a_0 .
 - Go through them in that order. Mark x *reachable* if $F(a_0, x)$, or if some earlier reachable y has $F(y, x)$, and remember y .
 - As soon as a reachable x also has $F(x, a_0)$, the chain wraps the whole circle. Follow the remembered links back to a_0 and output the chain.
5. If no starting lecture works, output “no valid subset exists”.

4 Why it is correct

Any chain found is valid. Let the chain be $a_0 = x_0, \dots, x_m$ with $F(x_i, x_{i+1})$ and $F(x_m, a_0)$ and increasing offsets. (Two different lectures with the same start overlap, so F never links them, and the offsets strictly increase.) Put each x_i on a line starting at a_0 : it occupies $[p_i, p_i + \ell_i)$ where p_i is its offset. Because x_{i+1} does not start inside x_i , we get $p_i + \ell_i \leq p_{i+1}$. Likewise x_m must not run past the start of a_0 , or it would overlap a_0 , which condition 1 of $F(x_m, a_0)$ forbids. So the arcs are disjoint, in the order $x_0, \dots, x_m$, and their neighbours are exactly the pairs we checked. By the claim, the set is valid.

No valid set is missed. Suppose a valid S exists. If $|S| = 1$, step 2 finds a lecture that works. Otherwise some member a_0 of S overlaps lecture 1, so it is tried in step 4. Listing S clockwise from a_0 gives a chain with increasing offsets, and F holds on all neighbouring pairs by the claim. The reachability search explores all such chains, so it finds one (maybe a different valid one), for this a_0 or an earlier one.

5 Complexity

- Overlap table: $O(n^2)$.
- Table F : n^2 pairs, $O(n)$ each, so $O(n^3)$.
- One chain search: sorting is $O(n \log n)$, and the reachability loop is $O(n^2)$ lookups. At most n starting lectures gives $O(n^3)$.

Total: $O(n^3)$ time and $O(n^2)$ space for the two tables. The first brute-force version tried every subset, which is $O(2^n)$. I did not try to prove $O(n^3)$ is optimal; it is just the easiest bound to justify.

6 Testing

The repository has 11 hand-checked test cases in `tests/all_testcases.txt`. They cover a single lecture, one lecture covering all, identical lectures, disjoint lectures, separate groups, a chain with a unique answer, touching lectures, midnight wrap-around, and rings of 4, 5 and 6 lectures. A ring where each lecture overlaps only its two neighbours has a valid subset exactly when its size is divisible by 3, which is why the rings of 4 and 5 have no answer and the ring of 6 does.